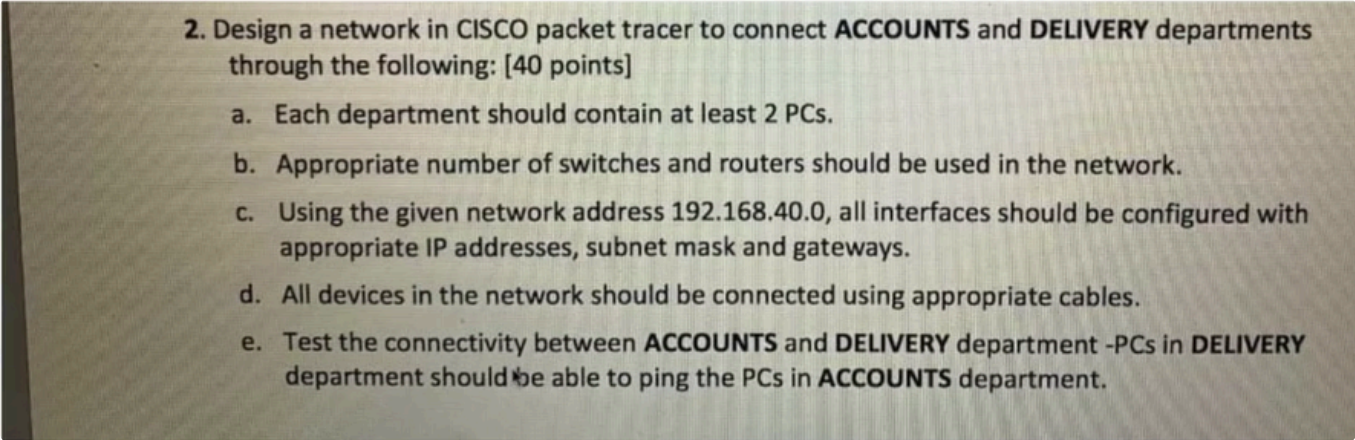


Simple Networking between two departments

Case Study

- 
2. Design a network in CISCO packet tracer to connect **ACCOUNTS** and **DELIVERY** departments through the following: [40 points]
- a. Each department should contain at least 2 PCs.
 - b. Appropriate number of switches and routers should be used in the network.
 - c. Using the given network address 192.168.40.0, all interfaces should be configured with appropriate IP addresses, subnet mask and gateways.
 - d. All devices in the network should be connected using appropriate cables.
 - e. Test the connectivity between **ACCOUNTS** and **DELIVERY** department -PCs in **DELIVERY** department should be able to ping the PCs in **ACCOUNTS** department.

Network=192.168.40.0

Two Departments = Required two subnets

2^n =number of subnets

N=1 (number of bit to borrow) , $2^1=2$ subnets

11111111.11111111.11111111.10000000

255.255.255.128

128=128 block size

Subnet 1

Network=192.168.40.0

Address Range = 192.168.40.1 - 192.168.40.126

Subnet Mask = 255.255.255.128

Broadcast Address = 192.168.40.127

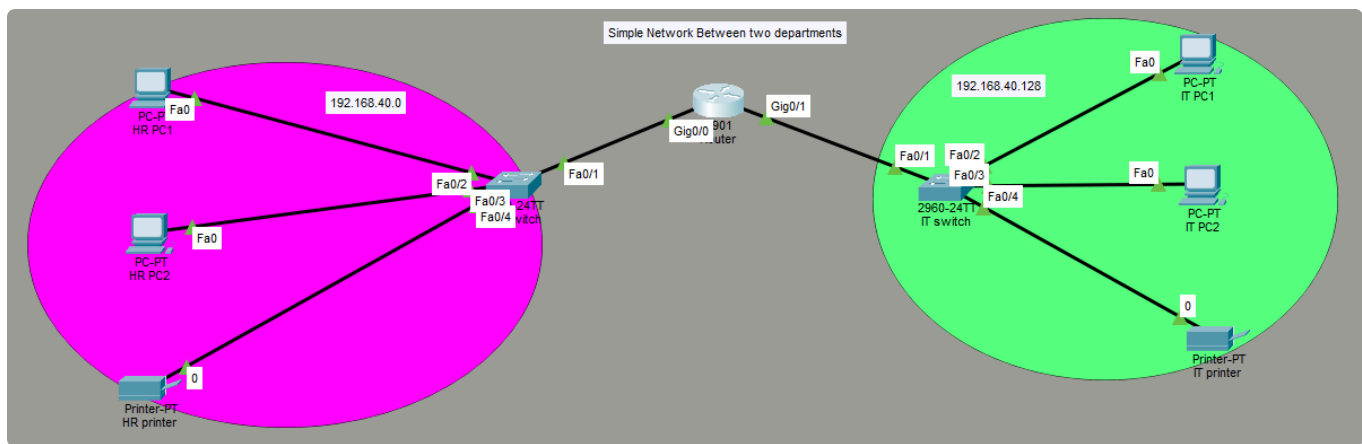
Subnet 2

Network = 192.168.40.128

Address Range = 192.168.40.129 - 192.168.40.254

Subnet Mask = 255.255.255.128

Broadcast Address = 192.168.40.255



```
Router(config)#interface gig0/1 Router(config-if)#ip address 192.168.40.129
255.255.255.128 Router(config-if)#no shut`
```

```
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.40.1 255.255.255.128
Router(config-if)#no shut
```

HR PC1

IPv4 Address:192.168.40.2

Subnet Mask:255.255.255.128

Default Gateway:192.168.40.1

HR PC2

IPv4 Address:192.168.40.3

Subnet Mask:255.255.255.128

Default Gateway:192.168.40.1

HR Printer

IPv4 Address:192.168.40.4

Default Gateway:255.255.255.128

IT PC1

IPv4 Address:192.168.40.130

Subnet Mask:255.255.255.128

Default Gateway:192.168.40.129

IT PC2

IPv4 Address:192.168.40.131

Subnet Mask:255.255.255.128

Default Gateway:192.168.40.129

HR Printer

IPv4 Address:192.168.40.132

Default Gateway:255.255.255.128